

Molly is buying lunchmeat and cheese from the Publix deli. Lunchmeat is \$5 per pound and cheese is \$4 per pound. Molly's mother gave her \$30 to spend. To determine the combinations of meat and cheese that she can afford to purchase, Molly wrote the inequality $5x + 4y \leq 30$.

1. Define the variables Molly used based on the context.

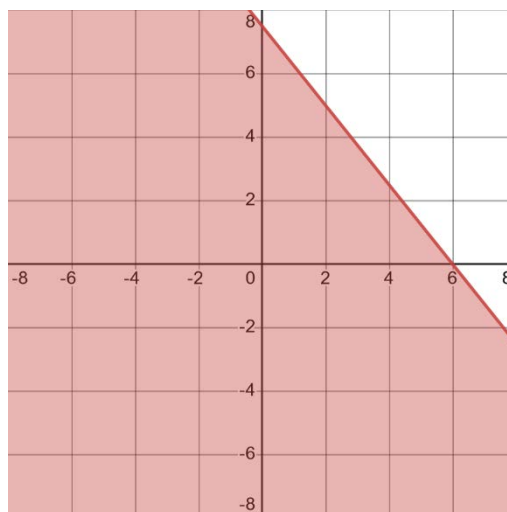
Let x represent lbs. of lunchmeat.

Let y represent lbs. of cheese.

2. Explain why the inequality sign $\leq$ makes sense in this context.

She cannot spend more than \$30, so it must be less than or equal to \$30.

3. Graph the inequality $5x + 4y \leq 30$.



4. The ordered pair $(5, -5)$ is a solution to the inequality $5x + 4y \leq 30$. Explain why this is not a viable solution in the context of buying lunchmeat and cheese.

You cannot buy a negative amount of anything.

5. The ordered pair $(2, 5)$ is a point on the boundary line. Explain what the ordered pair represents in the situation.

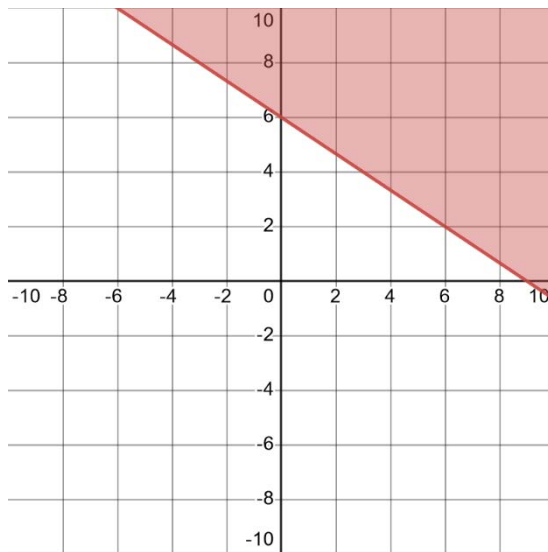
2 lbs of lunchmeat and 5 lbs of cheese can be bought for exactly \$30.

Talia invested some of her savings in the stock market. She bought 50 shares of a video streaming company and 75 shares of a manufacturing company. She hopes to make a combined profit of at least \$450 when she sells her shares. She wrote the inequality $50x + 75y \geq 450$ to represent the situation, where x is the change in value of one share of the streaming company and y is the change in value of one share of the manufacturing company.

6. Explain what it would mean if x or y were a negative number.

The value decreased.

7. Graph the inequality $50x + 75y \geq 450$.



8. Explain what the point $(14.75, -3.25)$ represents in Talia's situation.

The video streaming value increased by \$14.75 and the manufacturing company decreased by \$3.25.

9. Explain why it makes sense in Talia's context that there are no solutions for which both x and y are both negative.

Because if both decreased, she can't have a profit.

10. Stock prices in the United States are rounded to the nearest hundredths place. Complete the statement:

In order for solutions to the mathematical inequality to be viable solutions in the real-world context, the x and y values must be rounded to 2 decimal places.